

BIBEKANANDA BESRA

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EDUCATIONAL QUALIFICATIONS:

Masters of Computer Application (M.C.A)

Pondicherry University

May 2024 - May 2026(*expected*)

6.7 CGPA

Bachelors of Computer Applications (B.C.A)

Bangalore University

August 2019 - August 2022

7.85 CGPA

Higher Secondary (PCM/IT)

FM Higher Secondary school,

CHSE, Odisha

TECHNICAL SKILLS:

Programming Languages: Java, Python, JavaScript

Frontend: React.js, HTML5, CSS3

Backend & APIs: REST APIs, Node.js

DevOps & Infrastructure: Docker, Nginx, Proxmox, Linux, Cloudflare

Automation & Workflow: n8n, shell scripting

Databases: MySQL, MongoDB

Version Control: Git

Networking: VLAN configuration, DHCP setup, subnetting, NAS architecture

Operating Systems: Linux administration, Windows

PROJECT EXPERIENCE

Home Server Infrastructure (Proxmox Virtualization & AI Workloads)

Deployed and maintained a home server using **Proxmox VE** on a repurposed Lenovo laptop.

Created and managed multiple **Virtual Machines** and **LXC containers** for experimentation and service hosting.

Configured **GPU passthrough** for AI inferencing workloads using **Ollama**.

Implemented secure SSH access, system hardening, and routine security maintenance.

Managed storage allocation, networking, and virtualization lifecycle.

Technologies: Proxmox, Linux, Ollama, Virtualization, GPU Passthrough, SSH, Networking

Workflow Automation using n8n

This workflow helped me automate my blogging task.

Installed and configured **n8n** for workflow automation in a self-hosted environment.

Built automation pipelines to generate and publish blog content to **Ghost CMS**.

Integrated **Gemini Flash 2.0 API** for AI-assisted content generation.

Configured secure webhook triggers and environment variables.

Technologies: n8n, REST APIs, Ghost CMS, Docker, AI API Integration

Cloud-Deployed Docker Infrastructure on Azure

Provisioned and managed an **Azure VPS** to host containerized applications.

Deployed services including **Ghost CMS**, **n8n**, and **MongoDB** using Docker.

Handled container permission configurations, environment variables, and service networking.

Troubleshooting container execution issues and optimized resource usage.

Technologies: Azure VPS, Docker, MongoDB, Linux Server Administration

Lenovo Conservation Mode GNOME Extension (GNOME 46)

Developed a custom **GNOME Shell extension** to control Lenovo firmware-level battery conservation mode.

Built a privileged firmware interface with **Polkit-based authorization escalation**.

Integrated functionality into the **GNOME 46 SystemIndicator / Quick Settings architecture**.

Implemented structured UI components following GNOME extension best practices.

Technologies: JavaScript (GJS), GNOME 46, Polkit, Linux System Integration

Personal Portfolio Website

Designed and developed a responsive personal portfolio using **HTML**, **CSS**, and **JavaScript**.

Implemented SEO best practices including metadata optimization and performance tuning.

Focused on clean UI structure, semantic HTML, and fast loading performance.

Deployed publicly with domain integration.

Technologies: HTML5, CSS3, JavaScript, ReactJS, MongoDB, SEO Optimization

CERTIFICATIONS

1. Applied Accelerated Artificial Intelligence from Indian Institute of Technology, Guwahati 2025
2. Computer Vision and Image Processing from Indian Institute of Technology, Guwahati 2026(ongoing)